

Heart Disease Prediction using Artificial Neural Networks

Abstract

Heart disease is one of the leading causes of death worldwide. Early prediction and diagnosis can significantly reduce mortality rates. In this paper, an Artificial Neural Network (ANN) model is used to predict the presence of heart disease sedimentation medical attributes. The model is trained on a heart disease data-set containing features such as age, cholesterol, and blood pressure. The proposed model achieves high accuracy and is compared with the K-Nearest Neighbors (KNN) algorithm. Experimental results show that ANN performs better in terms of accuracy and reliability. This work demonstrates the effectiveness of machine learning techniques in healthcare prediction systems.

1.Introduction

Heart disease is a major health concern globally and requires early detection to prevent severe complications. Traditional diagnostic methods rely heavily on medical expertise and can be time-consuming. With the advancement of machine learning, automated systems can assist in predicting diseases using historical data. Artificial Neural

Networks (ANN) are powerful models inspired by the human brain and are widely used for classification problems. They can learn complex patterns from data and provide accurate predictions. This paper focuses on building an ANN-based model to predict heart disease and comparing its performance with a traditional machine learning algorithm.

1. Literature

Review Various studies have applied machine learning techniques for heart disease prediction. Traditional algorithms such as Decision Trees, Support Vector Machines, and K-Nearest Neighbors have been widely used. Recent research shows that ANN provides better accuracy due to its ability to model non-linear relationships. Neural networks can process large datasets efficiently and improve prediction performance. However, model tuning and data pre-processing play an important role in achieving better results.

3.Methodology

3.1 Datasets Description

The dataset used in this study contains medical records of patients. It includes attributes such as

- ✓ Age
- ✓ Sex
- ✓ Chest pain type (cp)
- ✓ Resting blood pressure (trestbps)
Cholesterol (chol)
- ✓ Maximum heart rate (thalach)
- ✓ The target variable indicates:
- ✓ 0 → No heart disease
- ✓ 1 → Presence of heart disease

3.2 Data Preprocessing

Data preprocessing is an important step before training the model. The following steps were performed:

- ✓ Handling missing values
- ✓ Feature selection
- ✓ Data normalization using Standard Scaler
- ✓ Splitting data set into training (80%) and testing (20%)

3.3 Artificial Neural Network Model

- ✓ The ANN model consists of:
- ✓ Input layer (based on number of features)

- ✓ Hidden layers with ReLU activation
- ✓ Output layer with Sigmoid activation
- ✓ The model is compiled using:
Optimizer: Adam
- ✓ Loss Function: Binary Crossentropy
- ✓ Evaluation Metric: Accuracy

4 .KNN Model (Extension)

To compare performance, the K-Nearest Neighbors algorithm is also implemented. KNN classifies data points based on the nearest neighbors and is simple yet effective for classification tasks.

4.Results and Discussion

The ANN model is trained for multiple epochs and evaluated on test data. The results show:

ANN Accuracy: ~95%

KNN Accuracy: ~95%

The confusion matrix indicates that most predictions are correct with very few misclassifications. The model performs well on unseen data.

The ANN model performs better due to its ability to learn complex patterns in data, whereas KNN is limited by distance-based classification.

5.Conclusion

This paper presents a heart disease prediction system using Artificial Neural Networks. The model achieves good accuracy and outperforms the KNN algorithm. The results demonstrate that ANN is effective for medical prediction tasks.

Future work includes:

- ✓ Using larger datasets

Applying deep learning models

Improving accuracy through hyper parameter tuning

References

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